

COVERAGE EXCHANGE

An underwriter stakes capital behind a claim about another chain's history. A lender releases credit only while the claim holds. Anyone who can prove one contradicting transaction takes the stake.

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CONTRACTS VERIFIED

Creditcoin CC3 testnet, chain 102031

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ATTACKS REFUSED

run against the live deployment

71

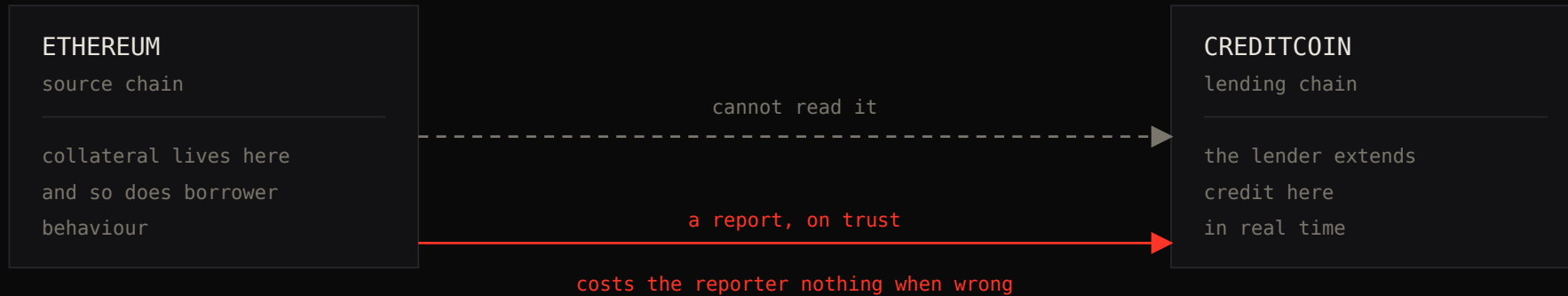
CONTRACT TESTS

across 9 suites, 0 failing

Live coverage-exchange.vercel.app

Repo github.com/subheeksh5599/coverage-exchange

Cross-chain credit runs on trust

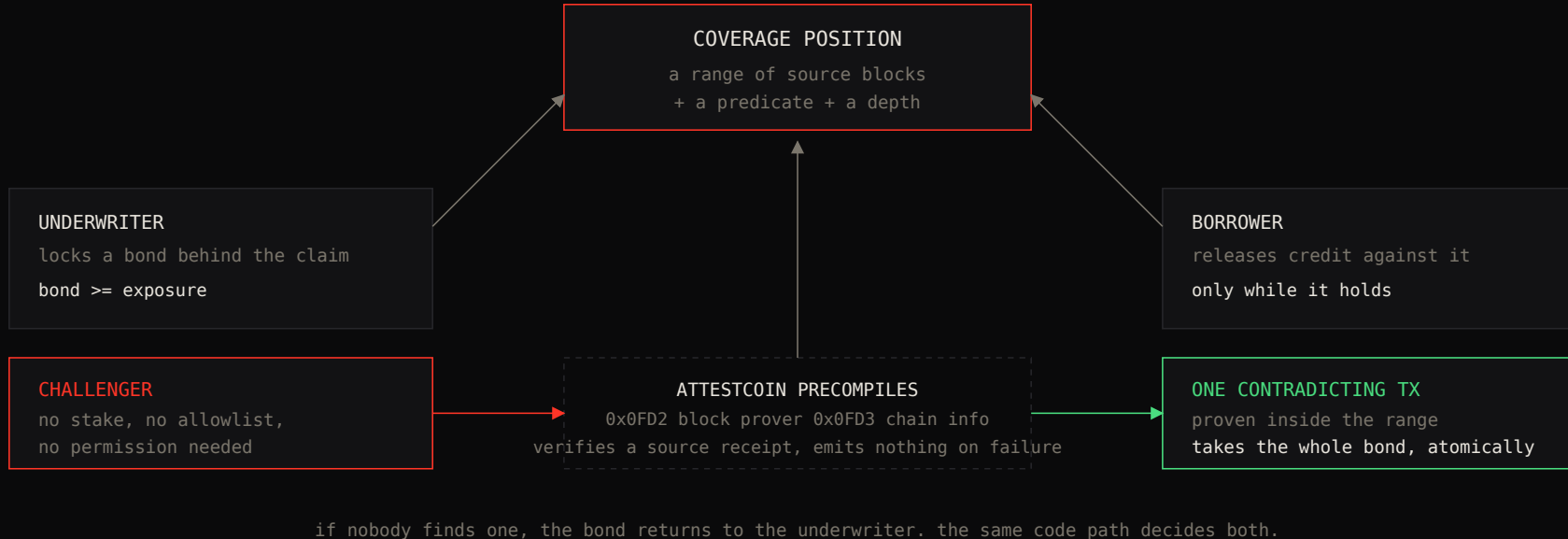


so the lender either believes the report, or lends so little it does not matter

The collateral and the borrower's behaviour are on Ethereum. The credit is on Creditcoin. The lending chain cannot read the source chain.

So it takes a report, and a report costs the reporter nothing when it is wrong. The alternative is lending so little that the report stops mattering.

Replace the report with a bond



BOND ≥ EXPOSURE

The underwriter's stake always covers what could be drawn, so breaching on purpose never pays.

VALIDITY IS COMPUTED

No stored "valid" flag. Every call recomputes it from the attested frontier and whether a counterexample has been proven.

NO COMMITTEE

No dispute window, no keeper, no admin key. The prover precompile and the invariant decide, in one transaction.

What the contract actually verifies

A challenge submits a source transaction hash and a Merkle proof. The engine does not take the submitter's word for any of it.

REJECTED AT EACH STEP

1. The Block Prover precompile (0x0FD2) verifies the proof, or the call reverts.
2. The receipt is decoded and the transaction must have **succeeded**, not merely been included.
3. The emitting contract must equal the position's declared **sourceContract**.
4. The log's **topic0** must equal the declared **eventSignature**.
5. Only then does the predicate run against the decoded values.

FAIL-CLOSED BY DEFAULT

The ChainInfo precompile (0x0FD3) supplies the source chain height. When it cannot answer, the protocol treats coverage as **unproven** rather than assuming it is fine.

Seven keyless checks run against the real testnet precompiles give **7/7** passing.

The proof is the only credential. There is no allowlist and no stake required to submit one.

Not a mock. Running state, read now

Read from the deployed contracts a few seconds before this slide was built, not from a fixture.

COVERAGE / EXCHANGE

BONDED
COVERAGE
- CC3

0X70AE...95EC

ATTESTED FRONTIER
- SEPOLIA

11,695,980

CHAININFO PRICEDFILL - ATTESTCOIN

01 OVERVIEW

02 BUY COVERAGE

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05 POSITIONS

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OVERVIEW

Your live position in the protocol. Balances, capacity, coverage and bonds are read from the deployed contracts on every refresh – if a read fails, this page says so instead of showing a state number.

BUY COVERAGE

PROVIDE COVERAGE

POSITIONS 13 0 active · 7 breached · 3 expired · 3 settled	BONDED CAPACITY 2,927,000 EXTUSD deposited in the engine	DRAWABLE LIQUIDITY 531,500 EXTUSD the pool can release now	BOND / EXPOSURE FLOOR 100% grace 10,000 blocks
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01 · YOUR POSITION

HALLET 3,007,116 EXTUSD testnet faucet asset	FREE CAPACITY 2,902,000 EXTUSD backing new coverage	BOND LOCKED 25,000 EXTUSD securing live exposure	YOUR COVERAGE 0 0 wild of 2 bought · 2,000 max exposure · 500 gram
AS LENDER 0 EXTUSD pool liquidity you supplied	YOUR RISK AT RISK 0 EXTUSD 13 positions you underwrite	GAS 4.999 ETC native balance for transactions	ONBOARD GET 8,000 EXTUSD a public function on the testnet token – a real hint, not a queue

02 · YOUR COVERAGE POSITIONS

#12

13

POSITIONS

7 breached · 3 expired · 3 settled

100%

BOND / EXPOSURE FLOOR

enforced per position

2,927,000

UNDERWRITER CAPITAL

cxTUSD held in the engine

531,500

DRAWABLE LIQUIDITY

releasable now

Nine contracts on Creditcoin CC3 testnet, all verified on Blockscout. Every figure here is a contract read, and a failed read is reported rather than cached.

A position that died

ALL POSITIONS

13 ON THIS DEPLOYMENT

ALL

AS BORROWER

AS UNDERWRITER

#13

BREACHED

INVALID - STATUS_BREACHED

MAX EXPOSURE	BOND LOCKED	DRAWN	PREMIUM PAID
1,000.00	1,000.00	500.00 - 500.00 left	10.40
COVERED WINDOW - SEPOLIA	REQUIRED DEPTH	BORROWER	UNDERWRITER
11,671,130 - 11,671,230	5390 - frontier passed	0x70AE...d5ec	0x70AE...d5ec

breached by counterexample - challenge key 0x6288612a...81d741 - bond paid to the challenger, draws frozen forever

INSPECT

Position 13 covered Sepolia blocks 11,671,130 to 11,671,230 under a ProhibitedRecipient predicate, evaluated against logs from the declared source contract. A transfer to the prohibited address inside that window is a proof.

That transfer exists: a real 100 USDC transfer on Sepolia at block 11,671,180. Its receipt is the entire input to the challenge.

0x19c528d3175bfc9d7cd0b1b7285fa07113054c5585eb8dead195b977edbc88a1

The bond moved to the challenger in the same transaction that proved it, and draws are frozen permanently.

Fifteen ways in. None of them worked.

Each attempt below was run against the live deployment and reverted. Two of them are failed transactions on the explorer, not simulations.

draw against an unknown position
 draw as a different counterparty
 draw above the position maximum
 underwriter withdraws the locked bond
 create a position with bond below exposure
 stranger calls the module wiring
 replayed counterexample
 counterexample against a settled position
 counterexample with a lookalike emitter
 draw against a breached position
 challenge race between two challengers
 draw after the live window closed
 source transaction that reverted
 proof for a transaction outside the window
 draw exceeding remaining capacity

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REFUSED

verified on the live deployment

2

ON-CHAIN FAILURES

openable on Blockscout

The two broadcast refusals:

draw against a breached position

0xb62ce5ed691e7521c2c05430c1fc8c5069d42ff649dc0db7680056df8969fbfa

replayed counterexample

0x65a8bc7f34a90538964342617ca75cc097f7d1bc3cbdd1c6e650898cb9772be7

The refusal reasons are typed errors, not strings: `CoverageNotValid`, `BondBelowExposure`, `NotTheCounterparty`.

Numbers, with the tool that produced them

GAS PER OPERATION		TEST SUITE	
purchase	371,652	71 tests	9 suites
draw	288,092	failures	0
challenge	396,004	fuzzed invariants	I-01..I-05, I-16
settlement	181,412	BATCHED VERIFICATION	
refused draw	195,748		
refused replay	266,336	continuity saving, 5 claims	30.1%
		continuity saving, 10 claims	22.5%
A refused path still costs gas. That is the point of a bonded claim: being wrong is not free.		Measured, not estimated. Reported as a real percentage rather than an order of magnitude, because that is what the run showed.	

A full lifecycle run of the mechanism took 225 seconds across 16 transactions at a 15-second block time, and the deployment passed 19 / 19 verification checks.

VERIFY IT

Every claim in this deck is checkable in a browser

THE APP

`coverage-exchange.vercel.app`

Positions, the attack matrix, the live attested frontier and the event log, all read from the deployed contracts.

THE CONTRACTS

Engine

`0xca5e3b0076673cd64a1655221305e66f2056d2bb`

Market

`0xb4569dc8827a8e573f2bb34b7f68a7ecf078b0e1`

Challenge manager

`0x5218279fd26e9b544c27e21acb1bfc9e325a5937`

All nine resolve on `creditcoin-testnet.blockscout.com`

OPEN THESE THREE

the breach

`blockscout.com/tx/`

`0x2f86470544c839fc43a25c31b8775299`

`7177ff28240c785477dfc17e5db70647`

the refused draw, status failed

`blockscout.com/tx/`

`0xb62ce5ed691e7521c2c05430c1fc8c50`

`69d42ff649dc0db7680056df8969fbfa`

the Sepolia proof source

`sepolia.etherscan.io/tx/`

`0x19c528d3175bfc9d7cd0b1b7285fa071`

`13054c5585eb8dead195b977edbc88a1`

The repo carries the contracts, the test suite, the evidence manifest the site is generated from, and the scripts that produced every transaction above.